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Regio- and stereospecific synthesis of polysubstituted alkenes by carbozincation of acetylenic sulfones

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Tri- or tetrasubstituted alkenes with various substituents can be constructed regio- and stereospecifically by treatment of acetylenic sulfones with organozinc reagents in tetrahydrofuran followed by hydrolysis or coupling with halohydrocarbon. Copyright © 2009 John Wiley & Sons, Ltd.

Keywords: acetylenic sulfone; organozinc reagents; carbozincation; alkenes; halohydrocarbon

Introduction

Polysubstituted alkenes are key structural elements of many natural products and pharmaceuticals.[1–5] Polysubstituted alkenes can also be used as intermediates in asymmetric transformation that generate quaternary centers such as osmylations, epoxidations and conjugate additions.[6] The efficient synthesis of polysubstituted alkenes in a regio- and stereoselective fashion is an important goal in organic chemistry.[7–10] Carbometallation of alkynes is a time-tested technique for the synthesis of substituted alkenes since the resulting alkenylmetals can be transformed to various polysubstituted alkenes stereoselectively.[11–16]

Organozinc compounds are very useful and versatile reagents for a variety of transformations in organic synthesis for the reason that organozinc reagents have distinct advantages compared with other organometallic reagents because they show high tolerance for functional groups.[17–21] Knochel *et al*. reported nickel-catalyzed carbozincation of alkynes and its application in organic synthesis.[22–24] Recently, Fu *et al*. reported Ni(II)-catalyzed asymmetric Negishi reactions of secondary α-bromoamides, secondary allylic or benzyl halides with organozinc reagents.[25,26] Marek *et al*. and Tanaka *et al*. reported copper-catalyzed carbozincation of alkyne.[27–29] However, carbozincation of alkynes and trapping the *in situ* formed alkenylzinc reagent with electrophiles other than a proton has received less attention.[22,28] In a preliminary communication, we have reported the one-pot three-component reaction of allylzinc bromide, acetylenic sulfone and halohydrocarbon.[30] As an extension of this research, we further explored carbozincation of acetylenic sulfones with various organozinc reagents, hoping to prepare polysubstituted alkenes bearing various substituents. Herein, we wish to report our detailed examination of the carbozincation of acetylenic sulfones followed by hydrolysis or trapping with halohydrocarbon, providing a regio- and stereospecific synthesis of differently substituted alkenes.

Results and Discussion

Firstly, the carbozincation of acetylenic sulfones followed by hydrolysis was investigated. We previously reported that the allylzincation products could be obtained in as high as 96% yield when 1.5 equiv of allylzinc bromide reacted with acetylenic sulfones in refluxing THF.[30] Considering the well-documented allylzincation reaction, the addition reaction of various organozinc reagents to acetylenic sulfone was conducted in similar reaction conditions. It was found that different organozinc halides can react with acetylenic sulfones smoothly. The reaction is general and the expected trisubstituted alkenes **3** can be obtained in good yields after hydrolysis. The typical results are summarized in Table 1. The R in acetylenic sulfones can be phenyl or pentyl. R^1 in organozinc halides can be allyl, benzyl, *n*-butyl or aryl. It should be noted that the carbozincation reaction of benzylzinc bromide, *n*-butylzinc chloride or arylzinc chloride with acetylenic sulfone completed in a shorter time in the presence of 10 mol% CuI (entries 3, 5 vs entries 4, 6, Table 1). This may be due to the fact that these organozinc reagents are less reactive than allylzinc bromide.[31]

The stereochemistry of the carbozincation was confirmed by the NOESY spectra of **3a** and **3b**. NOE enhancements were observed between the vinyl proton and methylene of the allyl group, indicating a *cis* relationship between the vinyl proton and allyl group. The molecular structure of

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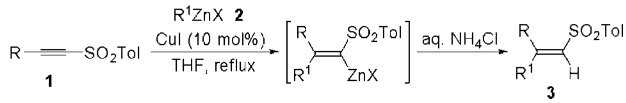
Table 1. The addition reaction of organozinc reagent to acetylenic sulfone



Entry	R	R^1ZnX	Time (h)	Yield (%)[a]
1	C_6H_5-	$CH_2{=}CHCH_2ZnBr$	2	**3a** 96[b]
2	*n*-C_5H_{11}-	$CH_2{=}CHCH_2ZnBr$	2	**3b** 92[b]
3	C_6H_5-	$PhCH_2ZnBr$	24	**3c** 90[b]
4	C_6H_5-	$PhCH_2ZnBr$	4	**3c** 89
5	*n*-C_5H_{11}-	$PhCH_2ZnBr$	12	**3d** 88[b]
6	*n*-C_5H_{11}-	$PhCH_2ZnBr$	2	**3d** 91
7	C_6H_5-	*n*-C_4H_9ZnCl	4	**3e** 91
8	C_6H_5-	C_6H_5ZnCl	4	**3f** 87
9	*n*-C_5H_{11}-	C_6H_5ZnCl	4	**3g** 89
10	C_6H_5-	*p*-FC_6H_4ZnCl	2	**3h** 85

[a] Isolated yield based on **1**. [b] No CuI was added.

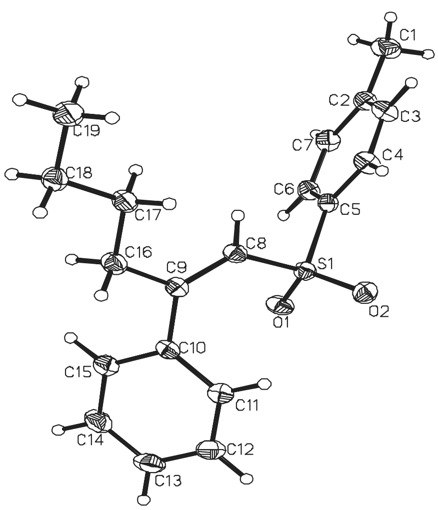


Figure 1. The molecular structure of **3e**.

compound **3e** was confirmed by X-ray diffraction analysis (Fig. 1), which also indicates that the carbozincation is in *syn*-fashion.

The CCDC deposition number for compound **3e** is 699216; crystal data, $C_{19}H_{22}O_2S$, MW = 314.43, monoclinic, space group *P*21/*c*, $a = 9.7904(8)$, $b = 17.0370(14)$, $c = 10.9707(9)$ Å; $\beta = 114.9660(10)^\circ$. $V = 1658.9(2)$ Å^3, $T = 293$ K, $Z = 4$, $D_c = 1.259$ g cm^{-1}, $\mu = 0.200$ mm^{-1}, $\lambda = 0.71073$ Å; *F*(000) 672, 3806 independent reflections ($R_{int} = 0.0471$), 14 270 reflections collected; refinement method, full-matrix least-squares on F^2; goodness-of-fit on $F^2 = 1.014$; final *R* indices (all data) $R_1 = 0.0826$, $wR_2 = 0.1233$.

One-pot multicomponent reaction has been of great interest for organic synthesis because it offers a convenient and economical method to prepare desired organic molecules.[32–36] On the basis of the above results, we further investigated the one-pot carbozincation–Negishi coupling reaction of acetylenic sulfones, organozinc halides and halohydrocarbons for the sake

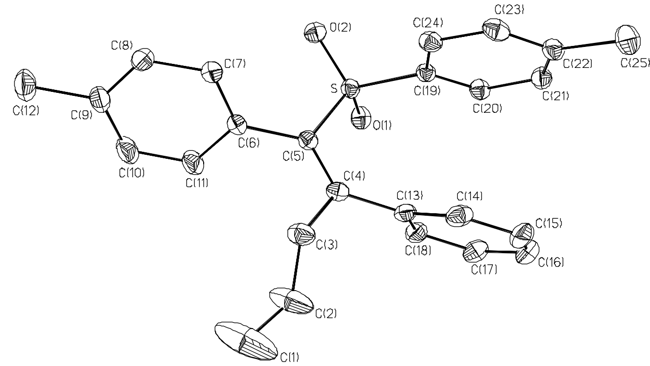


Figure 2. The molecular structure of **4ai**.

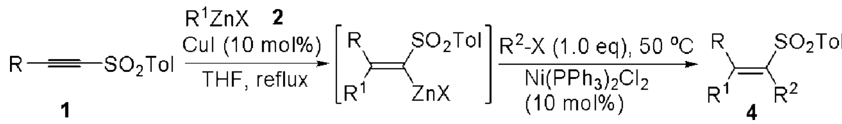
of providing an efficient method to prepare tetrasubstituted alkenes.

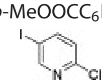
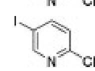
At 0 °C, acetylenic sulfones were added to the THF solution of RZnX. The reaction mixture was stirred at refluxing temperature for 2 h. After the carbozincation reaction was complete, halohydrocarbon, $NiCl_2(PPh_3)_2$ and THF were added successively at 50 °C to the reaction flask. After stirring at 50 °C for 12h, the desired products **4** were obtained. The results are shown in Table 2. The scope of the coupling reaction is quite general. These *in situ*-generated α-sulfonyl alkenylzinc intermediates readily participate in the Ni-catalyzed Negishi coupling reaction with halohydrocarbons to produce various tetrasubstituted alkenes. Table 2 indicates that R in acetylenic sulfones may be phenyl or *n*-pentyl; the organozinc halides may be allylzinc bromide, benzylzinc bromide, *n*-butylzinc chloride or *p*-fluorophenylzinc chloride; the halohydrocarbons may be allyl bromide, benzyl bromide, aryl bromide or aryl iodide. As demonstrated with various substrates, the procedure is highly tolerant to functional groups such as carbonyl group (entries 1, 2, 13, 14, 21 and 22, Table 2), methoxy group (entries 10 and 11, Table 2), halogen (entries 6, 7, 19 and 20, Table 2), nitrile group (entries 15, 16, 28 and 29, Table 2) and ester group (entries 17 and 18, Table 2). Although the yields were moderate, two kinds of substituents were introduced in a one-pot reaction. Since the one-pot three-component reactions proceeds regio- and stereospecifically, it would be applied to regioselective synthesis of tetrasubstituted alkenes by changing the combination of the organozinc reagent and halohydrocarbon (e.g. **4ae** vs **4bc; 4ag** vs **4da**).

The structures of tetrasubstituted alkenes **4ai, 4bc** and **4da** were affirmatively characterized by the X-ray diffraction analyses (Figs 2–4). The X-ray molecular structures of compounds **4ai, 4bc** and **4da** show that the double bond is in *Z*-configuration, which suggests that the one-pot carbozincation–Negishi coupling reaction of acetylenic sulfone, organozinc reagent and halohydrocarbon is in a *syn*-fashion. No anti-carbozincation–Negishi coupling products were obtained, indicating an excellent regio- and stereoselectivity of the one-pot three-component reaction.

The CCDC deposition number for compound **4ai** is 602635; cystal data, $C_{25}H_{24}O_2S$, MW = 388.50, triclinic, space group *P*-1 $a = 8.5858(6)$, $b = 11.5065(8)$, $c = 11.7576(8)$ Å; $\alpha = 72.0620(10)^\circ$, $\beta = 82.8060(10)^\circ$, $\gamma = 72.6040(10)^\circ$. $V = 1053.85(13)$ Å^3, $T = 293$ K, $Z = 2$, $D_c = 1.224$ g cm^{-1}, $\mu = 0.171$ mm^{-1}, $\lambda = 0.71073$ Å; F(000) 412, 3684 independent reflections ($R_{int} = 0.0213$), 5517 reflections collected; refinement method, full-matrix least-squares on F^2; goodness-

Table 2. One-pot three-component reaction of acetylenic sulfone, organozinc reagent and halohydrocarbon



Entry	R	R^1ZnX[b]	R^2X	Yield (%)[a]
1	C_6H_5-	$CH_2{=}CHCH_2ZnBr$ (**2a**)	*p*-$OHCC_6H_4Br$	**4aa** (63)
2	*n*-C_5H_{11}-	**2a**	*p*-$OHCC_6H_4Br$	**4ab** (60)
3	C_6H_5-	**2a**	$CH_2{=}CHCH_2Br$	**4ac** (83)
4	*n*-C_5H_{11}-	**2a**	$CH_2{=}CHCH_2Br$	**4ad** (74)
5	C_6H_5-	**2a**	$C_6H_5CH_2Br$	**4ae** (40)
6	C_6H_5-	**2a**	*p*-$BrC_6H_4CH_2Br$	**4af** (41)
7	C_6H_5-	**2a**	*p*-FC_6H_4I	**4ag** (61)
8	C_6H_5-	**2a**	C_6H_5Br	**4ah** (47)
9	C_6H_5-	**2a**	*p*-MeC_6H_4Br	**4ai** (60)
10	C_6H_5-	**2a**	*p*-$MeOC_6H_4I$	**4aj** (55)
11	*n*-C_5H_{11}-	**2a**	*p*-$MeOC_6H_4I$	**4ak** (45)
12	C_6H_5-	**2a**	*p*-$Me_2NC_6H_4Br$	**4al** (63)
13	C_6H_5-	**2a**	*p*-$MeCOC_6H_4I$	**4am** (61)
14	*n*-C_5H_{11}-	**2a**	*p*-$MeCOC_6H_4I$	**4an** (51)
15	C_6H_5-	**2a**	*p*-NCC_6H_4Br	**4ao** (57)
16	*n*-C_5H_{11}-	**2a**	*p*-NCC_6H_4Br	**4ap** (57)
17	C_6H_5-	**2a**	*p*-$MeOOCC_6H_4I$	**4aq** (40)
18	*n*-C_5H_{11}-	**2a**	*p*-$MeOOCC_6H_4I$	**4ar** (43)
19	C_6H_5-	**2a**		**4as** (51)
20	*n*-C_5H_{11}-	**2a**		**4at** (65)
21	C_6H_5-	$C_6H_5CH_2ZnBr$ (**2b**)	*p*-$OHCC_6H_4Br$	**4ba** (48)
22	*n*-C_5H_{11}-	**2b**	*p*-$OHCC_6H_4Br$	**4bb** (57)
23	C_6H_5-	**2b**	$CH_2{=}CHCH_2Br$	**4bc** (76)
24	C_6H_5-	**2b**	$C_6H_5CH_2Br$	**4bd** (63)
25	*n*-C_5H_{11}-	**2b**	$C_6H_5CH_2Br$	**4be** (66)
26	C_6H_5-	*n*-C_4H_9ZnCl (**2c**)	$CH_2{=}CHCH_2Br$	**4ca** (59)
27	*n*-C_5H_{11}-	**2c**	$CH_2{=}CHCH_2Br$	**4cb** (69)
28	C_6H_5-	**2c**	*p*-NCC_6H_4Br	**4cc** (46)
29	*n*-C_5H_{11}-	**2c**	*p*-NCC_6H_4Br	**4cd** (62)
30	C_6H_5-	*p*-FC_6H_4ZnCl (**2d**)	$CH_2{=}CHCH_2Br$	**4da** (72)

[a] Isolated yield based on **1**. [b] In the case of **2a**, no CuI was added.

of-fit on $F^2 = 1.153$; final R indices (all data) $R_1 = 0.0791$, $wR_2 = 0.1898$.

The CCDC deposition number for compound **4bc** is 699218; crystal data, $C_{25}H_{24}O_2S$, MW = 388.50, monoclinic, space group $P21/c$, $a = 10.9061(2)$, $b = 18.1560(4)$, $c = 10.6758(2)$ Å; $\beta = 100.1800(10)^\circ$. $V = 2080.65(7)$ Å^3, $T = 293$ K, $Z = 4$, $D_c = 1.240$ g cm^{-1}, $\mu = 0.173$ mm^{-1}, $\lambda = 0.71073$ Å; $F(000)$ 824, 4769 independent reflections ($R_{int} = 0.0299$), 19 224 reflections collected; refinement method, full-matrix least-squares on F^2; goodness-of-fit on $F^2 = 1.034$; final R indices (all data) $R_1 = 0.0565$, $wR_2 = 0.117$.

The CCDC deposition number for compound **4da** is 698893; crystal data, $C_{24}H_{21}FO_2S$, MW = 392.45, monoclinic, space group $P21/c$, $a = 9.8991(20)$, $b = 9.6219(20)$, $c = 21.8129(15)$ Å; $\beta = 95.575(10)^\circ$. $V = 2067.8(6)$ Å^3, $T = 298$ K, $Z = 4$, $D_c = 1.254$ g cm^{-1}, $\mu = 0.181$ mm^{-1}, $\lambda = 0.71073$ Å; $F(000)$ 816, 3648 independent reflections ($R_{int} = 0.0240$), 15 257 reflections collected; refinement method, full-matrix least-squares on F^2; goodness-of-fit on $F^2 = 0.944$; final R indices (all data) $R_1 = 0.0743$, $wR_2 = 0.1876$.

Conclusions

In summary, we have developed a simple and efficient method to synthesize tri- and tetrasubstituted alkenes regio- and stereospecifically by the carbozincation of acetylenic sulfones followed by hydrolysis or coupling with halohydrocarbon. Regioisomer of tetrasubstituted alkenes could be synthesized by changing the combination of the organozinc reagent and halohydrocarbon. The method has the advantages of a wide range substrates tolerance, mild reaction conditions and an excellent regio- and stereoselectivity.

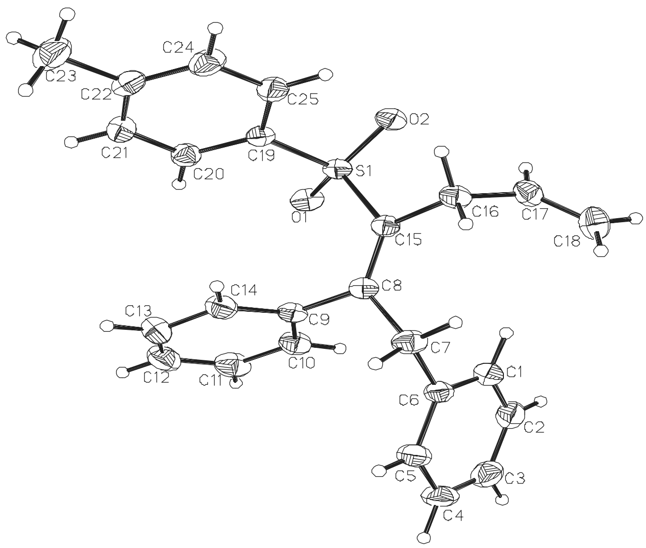


Figure 3. The molecular structure of **4bc**.

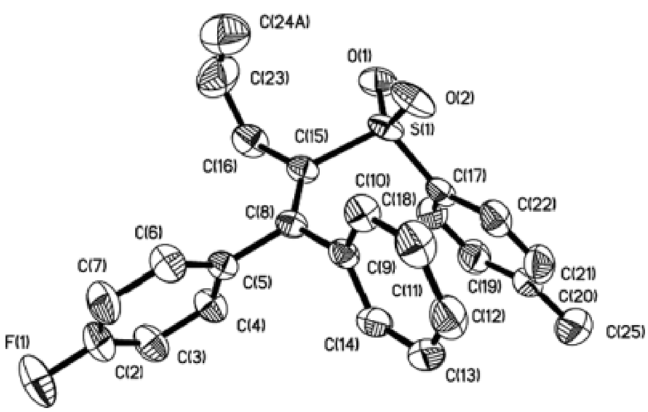


Figure 4. The molecular structure of **4da**.

Experimental

All solid products were recrystallized from ethyl acetate and hexane, and the melting points are uncorrected. All reactions were carried out under an argon atmosphere. THF was distilled from sodium–benzophenone. Halohydrocarbons in liquid state were purified by distillation before use. Halohydrocarbons in solid state were used without further purification. ^{1}H NMR spectra were measured at 300 MHz and ^{13}C NMR spectra were measured at 75 MHz in $CDCl_3$ or in C_6D_6 with TMS as the internal standard. Acetylenic sulfones[37] and organozinc reagents[38,39] were prepared according to literature procedures.

General procedure for the synthesis of trisubstituted alkenes 3a and 3b

The Schlenk flask with a condenser was charged with THF (1.4 ml), activated zinc dust (50 mg, 0.75 mmol) and allyl bromide (0.06 ml, 0.75 mmol) under argon atmosphere. The mixture was stirred at 0 °C until the zinc disappeared. Acetylenic sulfone (0.5 mmol) was added to the reaction mixture and stirred at refluxing temperature for 2 h. The mixture was quenched with a saturated NH_4Cl solution after the disappearance of acetylenic sulfone (monitored by TLC) and was extracted with ethyl acetate (3 × 10 ml). The organic layer was combined and dried over $MgSO_4$. After filtration and removal of solvent *in vacuo*, the crude product was purified with flash chromatography and the desired product **3a** or **3b** was obtained.

*(Z)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(3a)***

White solid; m.p. 64–65 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.35 (d, $J = 8.2$ Hz, 2H, Ar–*H*), 7.29–7.16 (m, 3H, Ar–*H*), 7.08 (d, $J = 8.1$ Hz, 2H, Ar–*H*), 7.02–6.99 (m, 2H, Ar–*H*), 6.46 (s, 1H, =C*H*), 5.72–5.59 (m, 1H, =C*H*), 5.11–5.01 (m, 2H, =CH_2), 3.05 (d, $J = 6.8$ Hz, 2H, CH_2), 2.33 (s, 3H, CH_3); ^{13}C NMR (75 MHz $CDCl_3$) δ 156.1 (=C), 143.8, 138.6, 136.8 (Ar–*C*), 132.4 (=CH), 129.8, 129.4, 128.4, 127.9, 127.7 (Ar–*C*), 127.6 (=CH), 119.5 (=CH_2), 44.7 (CH_2), 21.6 (CH_3); HRMS (ESI), calcd for $C_{18}H_{19}O_2S$ (MH^+), 299.1106, found, 299.1100; IR (KBr) 3080, 1738, 1596, 1313, 1140, 1086 cm^{-1}.

*(E)-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(3b)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.73 (d, $J = 8.1$ Hz, 2H, Ar–*H*), 7.26 (d, $J = 8.1$ Hz, 2H, Ar–*H*), 6.09 (s, 1H, =C*H*), 5.66–5.55 (m, 1H, =C*H*), 5.08–4.97 (m, 2H, =CH_2), 2.81 (d, $J = 6.6$ Hz, 2H, CH_2), 2.52 (t, $J = 7.8$ Hz, 2H, CH_2), 2.36 (s, 3H, CH_3), 1.33–1.22 (m, 6H, 3CH_2), 0.82 (t, $J = 6.6$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.3 (=C), 143.9, 139.6, 133.1 (Ar–*C*), 129.8 (=*C*H), 127.2 (=*C*H), 127.0 (Ar–*C*), 119.0 (=*C*H_2), 41.5 (CH_2), 31.9 (CH_2), 31.3 (CH_2), 27.8 (CH_2), 22.4 (CH_2), 21.6 (CH_3), 13.9 (CH_3); HRMS (ESI), calcd for $C_{17}H_{25}O_2S$ (MH^+), 293.1575, found, 293.1578; IR (KBr) 2955, 2929, 1619, 1597, 1466, 1301, 1145, 1086 cm^{-1}.

General procedure for the synthesis of trisubstituted alkenes 3c–3h

To the solution of RZnX (0.6 mmol) in THF (1.4 ml) was added acetylenic sulfone (0.5 mmol) and CuI (10 mol%, 0.06 mmol) at 0 °C. The reaction mixture was then stirred at refluxing temperature for 2 h. The mixture was quenched with a saturated NH_4Cl solution after the reaction was complete (monitored by TLC). After the usual workup, the crude product was purified with flash chromatography on silica gel to give the desired products **3c–3h**.

*(Z)-2,3-diphenyl-1-(p-tolylsulfonyl)-1-propylene **(3c)***

White solid; m.p. 97–98 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.39–7.36 (m, 2H, Ar–*H*), 7.28–7.19 (m, 6H, Ar–*H*), 7.13–7.06 (m, 4H, Ar–*H*), 6.94 (d, $J = 7.1$ Hz, 2H, Ar–*H*), 6.40 (s, 1H, =C*H*), 3.64 (s, 2H, CH_2), 2.37 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.1 (=*C*), 143.6, 136.6, 135.7, 130.7, 129.5, 129.3, 128.7, 128.3, 127.8, 127.7, 127.6, 127.2 (Ar–*C*), 127.0 (=*C*H), 47.1 (CH_2), 21.6 (CH_3); HRMS (ESI), calcd for $C_{22}H_{21}O_2S$ (MH^+), 349.1262, found, 349.1266; IR (KBr) 2921, 1596, 1492, 1312, 1301, 1137 cm^{-1}.

*(E)-2-benzyl-1-(p-tolylsulfonyl)-1-heptylene **(3d)***

White solid; m.p. 72–73 °C; ^{1}H NMR (300 MHz, C_6D_6) δ 7.90 (d, $J = 8.2$ Hz, 2H, Ar–*H*), 7.14–7.06 (m, 3H, Ar–*H*), 6.91–6.88 (m, 2H, Ar–*H*), 6.82 (d, $J = 8.0$ Hz, 2H, Ar–*H*), 6.19 (s, 1H, =C*H*), 3.07 (s, 2H, CH_2), 2.71 (t, $J = 7.9$ Hz, 2H, CH_2), 1.92 (s, 3H, CH_3), 1.48–1.43 (m, 2H, CH_2), 1.32–1.30 (m, 4H, 2CH_2), 0.92 (t, $J = 6.6$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, C_6D_6) δ 158.6 (=*C*), 143.2, 140.7, 137.0 (Ar–*C*), 129.2 (=*C*H), 128.8, 128.1, 127.8, 127.3, 126.9 (Ar–*C*), 43.6 (CH_2), 32.0 (CH_2), 30.7 (CH_2), 28.3 (CH_2), 22.5 (CH_2), 20.9 (CH_3), 13.9 (CH_3); HRMS (ESI), calcd for $C_{21}H_{26}NaO_2S$ (MNa^+), 365.1551, found, 365.1545; IR (KBr) 2912, 1595, 1491, 1353, 1120 cm^{-1}.

(Z)-2-phenyl-1-(p-tolylsulfonyl)-1-hexylene ***(3e)***

White solid; m.p. 117–118 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.37 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.33–7.23 (m, 3H, Ar–*H*), 7.11 (d, *J* = 8.0 Hz, 2H, Ar–*H*), 7.01 (d, *J* = 6.4 Hz, 2H, Ar–*H*), 6.50 (s, 1H, =C*H*), 2.36 (m, 5H, CH_3, CH_2), 1.33–1.21 (m, 4H, 2CH_2), 0.84 (t, *J* = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.9 (=*C*), 143.2, 138.3, 136.4, 128.8, 128.5, 127.8, 127.4, 127.3 (Ar–*C*), 127.2 (=*C*H), 40.3 (CH_2), 28.7 (CH_2), 21.7 (CH_2), 21.2 (CH_3), 13.4 (CH_3); HRMS (ESI), calcd for $C_{19}H_{22}NaO_2S$ (MNa^+), 337.1238, found, 337.1238; IR (KBr) 3055, 2872, 1613, 1597, 1349, 1145 cm^{-1}.

1,1-diphenyl-2-(p-tolylsulfonyl)ethylene ***(3f)***

White solid; m.p. 105–106 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.48 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.38–7.30 (m, 6H, Ar–*H*), 7.22–7.09 (m, 6H, Ar–*H*), 7.00 (s, 1H, =C*H*), 2.38 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 154.4 (=*C*), 143.4, 138.8, 138.2, 135.2, 129.9, 129.4, 129.0, 128.6, 128.5, 128.2, 127.9, 127.5 (Ar–*C*), 127.3 (=*C*H), 21.2 (CH_3); HRMS (ESI), calcd for $C_{21}H_{18}NaO_2S$ (MNa^+), 357.0925, found, 357.0921; IR (KBr) 3039, 1606, 1592, 1491, 1335, 1150, 1085 cm^{-1}.

(E)-2-phenyl-1-(p-tolylsulfonyl)-1-heptylene ***(3g)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.78 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.28–7.24 (m, 7H, Ar–*H*), 6.39 (s, 1H, =C*H*), 2.94 (t, *J* = 7.9 Hz, 2H, CH_2), 2.35 (s, 3H, CH_3), 1.26–1.06 (m, 6H, 3CH_2), 0.73 (t, *J* = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.8 (=*C*), 143.8, 139.2, 138.9, 129.5, 129.2, 128.4, 127.5, 126.9 (Ar–*C*), 126.3 (=*C*H), 31.4 (CH_2), 29.9 (CH_2), 27.8 (CH_2), 22.0 (CH_2), 21.3 (CH_3), 13.6 (CH_3); HRMS (ESI), calcd for $C_{20}H_{24}NaO_2S$ (MNa^+), 351.1395, found, 351.1396; IR (KBr) 3059, 2956, 1598, 1494, 1462, 1315, 1145, 1086 cm^{-1}.

(E)-1-(p-fluorophenyl)-1-phenyl-2-(p-tolylsulfonyl)ethylene ***(3h)***

White solid; m.p. 97–98 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.47 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.32–6.97 (m, 12H, Ar–*H*), 2.35 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 165.6, 162.3 (Ar–*C*), 153.5 (=*C*), 143.9, 138.5, 135.4, 131.9, 130.4, 129.7, 129.0, 128.8, 128.2 (Ar–*C*), 127.7 (=*C*H), 115.8 (Ar–*C*), 21.6 (CH_3); HRMS (ESI), calcd for $C_{21}H_{17}FNaO_2S$ (MNa^+), 375.0831, found, 375.0825; IR (KBr) 3054, 1597, 1491, 1336, 1142, 1118 cm^{-1}.

General procedure for the synthesis of tetrasubstituted alkenes 4

Alkenylzinc intermediates were prepared *in situ* according to the procedures described above. Once the carbozincation was complete, halohydrocarbon (0.5 mmol), $NiCl_2(PPh_3)_2$ (33 mg, 10% mol) and 3.0 ml of THF were added successively to the reaction flask. The mixture was stirred at 50 °C for 12 h. After the reaction was complete (monitored by TLC), the mixture was quenched with a saturated NH_4Cl solution. After usual workup, the crude product was purified with flash chromatography (silica/hexane–ethyl acetate 8 : 1 v/v) and the desired products **4** were obtained.

(Z)-1-(p-formylphenyl)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4aa)***

White solid; m.p. 104–106 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.06 (s, 1H, C*H*O), 7.88 (d, *J* = 8.4 Hz, 2H, Ar–*H*), 7.42 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.35–7.33 (m, 3H, Ar–*H*), 7.24–7.16 (m, 4H, Ar–*H*), 7.08–7.05 (m, 2H, Ar–*H*), 5.46–5.37 (m, 1H, =C*H*), 4.90 (dd, *J* = 1.5 Hz, 10.2 Hz, 1H, =CH_2), 4.68 (dd, *J* = 1.5 Hz, 17.1 Hz, 1H, =CH_2), 2.86–2.83 (m, 2H, CH_2), 2.35 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 191.8 (*C*HO), 151.8, 144.0, 141.0, 140.1 (Ar–*C*), 137.8 (=*C*H), 137.6 (=*C*), 136.2, 132.1, 132.0, 129.5, 129.2, 128.3, 128.0, 127.9 (Ar–*C*), 127.8 (=*C*), 118.3 (=CH_2), 43.6 (CH_2), 21.6 (CH_3); HRMS (ESI), calcd for $C_{25}H_{23}O_3S$ (MH^+), 403.1368, found, 403.1362; IR (KBr) 3051, 2835, 1698, 1601, 1568, 1441, 1302, 1149, 1086 cm^{-1}.

(E)-1-(p-formylphenyl)-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4ab)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.02 (s, 1H, C*H*O), 7.78 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.50 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.29–7.17 (m, 4H, Ar–*H*), 5.61–5.51 (m, 1H, =C*H*), 5.02 (d, *J* = 9.0 Hz, 1H, =CH_2), 4.88 (dd, *J* = 1.2 Hz, 16.8 Hz, 1H, =CH_2), 2.87 (t, *J* = 7.5 Hz, 2H, CH_2), 2.62 (d, *J* = 6.6 Hz, 2H, CH_2), 2.41 (s, 3H, CH_3), 1.63–1.51 (m, 2H, CH_2), 1.42–1.35 (m, 4H, 2CH_2), 0.96 (t, *J* = 6.90 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 191.9 (*C*HO), 154.7 (Ar–*C*), 144.2 (=*C*), 140.2, 138.0 (Ar–*C*), 135.8 (=*C*H), 133.5, 131.4, 129.6, 129.4, 129.3 (Ar–*C*), 128.1 (=*C*), 117.8 (=CH_2), 40.3 (CH_2), 32.2 (CH_2), 31.6 (CH_2), 29.1 (CH_2), 22.5 (CH_2), 21.6 (CH_3), 14.1 (CH_3); HRMS (ESI), calcd for $C_{24}H_{29}O_3S$ (MH^+), 397.1832, found, 397.1833; IR (KBr) 3027, 2947, 2813, 1697, 1604, 1567, 1494, 1301, 1146, 1085 cm^{-1}.

(Z)-4-phenyl-5-(p-tolylsulfonyl)-1,4,7-octatriene ***(4ac)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.21–7.11 (m, 5H, Ar–*H*), 7.02 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 6.87–6.84 (m, 2H, Ar–*H*), 6.04–5.91 (m, 1H, =C*H*), 5.62–5.49 (m, 1H, =C*H*), 5.27–5.18 (m, 2H, =CH_2), 5.02–4.92 (m, 2H, =CH_2), 3.50 (d, *J* = 5.9 Hz, 2H, CH_2), 3.13 (d, *J* = 6.6 Hz, 2H, CH_2), 2.34 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.0, 143.1, 139.0 (Ar–*C*), 138.7 (=*C*), 138.5 (=*C*H), 134.4 (=*C*H), 132.2, 129.0, 128.5, 127.7, 127.5 (Ar–*C*), 127.4 (=*C*), 117.9 (=CH_2), 116.8 (=CH_2), 42.2 (CH_2), 32.8 (CH_2), 21.5 (CH_3); HRMS (ESI), calcd for $C_{21}H_{23}O_2S$ (MH^+), 339.1419, found, 339.1413; IR (KBr) 3061, 2923, 1596, 1491, 1443, 1311, 1148 cm^{-1}.

(E)-4-pentyl-5-(p-tolylsulfonyl)-1,4,7-octatriene ***(4ad)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.64 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.18 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 5.72–5.52 (m, 2H, =C*H*), 5.04–4.84 (m, 4H, =CH_2), 3.10 (d, *J* = 5.7 Hz, 2H, CH_2), 2.80 (d, *J* = 6.1 Hz, 2H, CH_2), 2.40 (t, *J* = 8.6 Hz, 2H, CH_2), 2.30 (s, 3H, CH_3), 1.20–1.05 (m, 6H, 3CH_2), 0.74 (t, *J* = 6.3 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 154.3 (Ar–*C*), 143.7 (=*C*), 139.7 (=*C*H), 135.4 (=*C*H), 134.6, 133.4, 129.6 (Ar–*C*), 127.3 (=*C*), 117.3 (=CH_2), 116.3 (=CH_2), 38.3 (CH_2), 33.4 (CH_2), 33.2 (CH_2), 32.2 (CH_2), 28.7 (CH_2), 22.5 (CH_2), 21.6 (CH_3), 14.0 (CH_3); HRMS (ESI), calcd for $C_{20}H_{29}O_2S$ (MH^+), 333.1888, found, 333.1891; IR (KBr) 2928, 2870, 1636, 1596, 1493, 1301, 1145, 1017, 916 cm^{-1}.

(Z)-4,6-diphenyl-5-(p-tolylsulfonyl)-1,4-hexadiene ***(4ae)***

White solid; m.p. 108–109 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.36–7.34 (m, 4H, Ar–*H*), 7.31–7.15 (m, 4H, Ar–*H*), 7.06 (d, *J* = 8.3 Hz, 2H, Ar–*H*), 6.97–6.94 (m, 4H, Ar–*H*), 5.48–5.39 (m, 1H, =C*H*), 4.96–4.82 (m, 2H, =CH_2), 4.16 (s, 2H, CH_2), 3.16 (d, *J* = 6.5 Hz, 2H, CH_2), 2.31 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.7, 143.1 (Ar–*C*), 140.1 (=*C*), 138.6, 138.5 (Ar–*C*), 138.1 (=*C*H), 131.9, 128.8, 128.6, 128.5, 128.3, 127.7, 127.5 (Ar–*C*), 126.5 (=*C*), 117.9 (=CH_2), 42.8 (CH_2), 34.3 (CH_2), 21.5 (CH_3); HRMS (ESI), calcd for $C_{25}H_{24}NaO_2S$ (MNa^+), 411.1395, found, 411.1389; IR (KBr) 3061, 1597, 1494, 1453, 1301, 1145 cm^{-1}.

(Z)-6-(p-bromophenyl)-4-phenyl-5-(p-tolylsulfonyl)-1,4-hexadiene ***(4af)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.47 (d, J = 8.1 Hz, 2H, Ar–H), 7.26–7.15 (m, 5H, Ar–H), 7.07–6.91 (m, 6H, Ar–H), 5.46–5.38 (m, 1H, =CH), 4.99–4.83 (m, 2H, =CH_2), 4.10 (s, 2H, CH_2), 3.14 (d, J = 6.4 Hz, 2H, CH_2), 2.33 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.9, 143.3 (Ar–C), 139.8 (=C), 138.4, 138.3 (Ar–C), 137.2 (=CH), 131.7, 130.04, 128.9, 128.5, 127.7, 127.6, 127.5 (Ar–C), 120.4 (=C), 118.1 (=CH$_2$), 42.8 (CH$_2$), 33.8 (CH$_2$), 21.5 (CH$_3$); HRMS (ESI), calcd for $C_{25}H_{24}BrO_2S$ (MH$^+$), 467.0680, found, 467.0676; IR (KBr) 3060, 2946, 1595, 1487, 1441, 1301, 1147, 1082 cm^{-1}.

(Z)-1-(p-fluorophenyl)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4ag)***

White solid; m.p. 132–134 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.42–7.30 (m, 3H, Ar–H), 7.27–7.15 (m, 6H, Ar–H), 7.10–7.03 (m, 4H, Ar–H), 5.51–5.38 (m, 1H, =CH), 4.91 (d, J = 9.9 Hz, 1H, =CH_2), 4.72 (d, J = 17.1 Hz, 1H, =CH_2), 2.88 (d, J = 6.6 Hz, 2H, CH_2), 2.36 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 164.6, 161.3, 151.7, 143.7, 140.7 (Ar–C), 138.2 (=C), 137.6 (=CH), 133.1, 132.4, 129.5, 129.1, 128.3, 128.0 (Ar–C), 127.7 (=C), 118.0 (=CH$_2$), 115.3 (Ar–C), 43.7 (CH$_2$), 21.6 (CH$_3$); HRMS (ESI), calcd for $C_{24}H_{21}FNaO_2S$ (MNa$^+$), 415.1144, found, 415.1139; IR (KBr) 3052, 2980, 1623, 1506, 1319, 1146, 1087 cm^{-1}.

(Z)-1,2-diphenyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4ah)***

White solid; m.p. 142–143 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.37–7.33 (m, 6H, Ar–H), 7.23–7.18 (m, 6H, Ar–H), 7.07 (d, J = 8.1 Hz, 2H, Ar–H), 5.51–5.40 (m, 1H, =CH), 4.89 (d, J = 10.2 Hz, 1H, =CH_2), 4.70 (dd, J = 1.5 Hz, 16.9 Hz, 1H, =CH_2), 2.87 (d, J = 6.9 Hz, 2H, CH_2), 2.36 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.0, 143.5, 141.6, 138.4 (Ar–C), 137.8 (=CH), 133.6 (=C), 132.7, 131.2, 129.0, 128.7, 128.4, 128.2, 128.1, 127.7 (Ar–C), 127.6 (=C), 117.9 (=CH$_2$), 43.6 (CH$_2$), 21.6 (CH$_3$); HRMS (ESI), calcd for $C_{24}H_{23}O_2S$ (MH$^+$), 375.1419, found, 375.1413; IR (KBr) 3058, 1597, 1313, 1144 cm^{-1}.

(Z)-2-phenyl-1-(p-tolyl)-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4ai)***

White solid; m.p. 119–120 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.34–7.32 (m, 3H, Ar–H), 7.26 (d, J = 8.0 Hz, 2H, Ar–H), 7.16–7.06 (m, 6H, Ar–H), 7.09–6.99 (m, 2H, Ar–H), 5.52–5.38 (m, 1H, =CH), 4.90 (d, J = 9.9 Hz, 1H, =CH_2), 4.72 (d, J = 17.1 Hz, 1H, =CH_2), 2.89 (d, J = 6.9 Hz, 2H, CH_2), 2.39 (s, 3H, CH_3), 2.36 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.1, 143.4, 142.1, 138.6, 138.5 (Ar–C), 137.9 (=CH), 132.8 (=C), 131.1, 130.5, 129.0, 128.9, 128.4, 128.1 (Ar–C), 127.6 (=C), 117.9 (=CH$_2$), 43.7 (CH$_2$), 21.6 (CH$_3$), 21.4 (CH$_3$); HRMS (ESI), calcd for $C_{25}H_{25}O_2S$ (MH$^+$), 389.1575, found, 389.1569; IR (KBr) 3023, 1596, 1445, 1318, 1147 cm^{-1}.

(Z)-1-(p-methoxyphenyl)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4aj)***

White solid m.p. 129–130 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.37–7.35 (m, 3H, Ar–H), 7.28 (d, J = 8.2 Hz, 2H, Ar–H), 7.22–7.20 (m, 2H, Ar–H), 7.18–7.15 (m, 2H, Ar–H), 7.12–7.10 (m, 2H, Ar–H), 6.91 (d, J = 8.6 Hz, 2H, Ar–H), 5.55–5.41 (m, 1H, =CH), 4.93 (d, J = 9.9 Hz, 1H, =CH_2), 4.75 (d, J = 17.0 Hz, 1H, =CH_2), 3.87 (s, 3H, OCH_3), 2.92 (d, J = 6.2 Hz, 2H, CH_2), 2.39 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.8, 151.2, 143.4, 141.2, 138.6 (Ar–C), 138.0 (=CH), 132.8 (=C), 132.5, 129.0, 128.4, 128.0, 127.6 (Ar–C), 125.5 (=C), 117.8 (=CH$_2$), 113.7 (Ar–C), 55.3 (OCH$_3$), 43.7 (CH$_2$), 21.6 (CH$_3$); HRMS (ESI), calcd for $C_{25}H_{24}NaO_3S$ (MNa$^+$), 427.1344, found, 427.1338; IR (KBr) 3060, 1596, 1508, 1316, 1249, 1146 cm^{-1}.

(E)-1-(p-methoxyphenyl)-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4ak)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.48 (d, J = 8.2 Hz, 2H, Ar–H), 7.18 (d, J = 8.0 Hz, 2H, Ar–H), 6.88 (d, J = 8.7 Hz, 2H, Ar–H), 6.75 (d, J = 8.7 Hz, 2H, Ar–H), 5.61–5.52 (m, 1H, =CH), 4.98 (d, J = 10.1 Hz, 1H, =CH_2), 4.89 (dd, J = 1.3 Hz, 18.4 Hz, 1H, =CH_2), 3.79 (s, 3H, OCH_3), 2.84 (t, J = 8.0 Hz, 2H, CH_2), 2.64 (d, J = 6.5 Hz, 2H, CH_2), 2.39 (s, 3H, CH_3), 1.61–1.38 (m, 2H, CH_2), 1.41–1.39 (m, 4H, 2CH_2), 0.94 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.5, 154.1, 143.5, 138.7, 138.5 (Ar–C), 134.2 (=C), 132.5 (=CH), 129.1, 128.1 (Ar–C), 126.5 (=C), 117.3 (=CH$_2$), 113.4 (Ar–C), 55.2 (OCH$_3$), 40.3 (CH$_2$), 32.2 (CH$_2$), 31.5 (CH$_2$), 29.1 (CH$_2$), 22.5 (CH$_2$), 21.5 (CH$_3$), 14.0 (CH$_3$); HRMS (ESI), calcd for $C_{24}H_{31}O_3S$ (MH$^+$), 399.1984, found, 399.1997; IR (KBr) 2929, 2954, 2853, 1601, 1462, 1312, 1289, 1249, 1144, 1085 cm^{-1}.

(Z)-1-(p-N,N-dimethylaminophenyl)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4al)***

White solid; m.p. 168–170 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.32–7.30 (m, 3H, Ar–H), 7.24 (d, J = 8.2 Hz, 2H, Ar–H), 7.17–7.12 (m, 2H, Ar–H), 7.07–7.04 (m, 4H, Ar–H), 6.66 (d, J = 8.8 Hz, 2H, Ar–H), 5.51–5.36 (m, 1H, =CH), 4.88 (dd, J = 1.7 Hz, 10.1 Hz, 1H, =CH_2), 4.73 (dd, J = 1.6 Hz, 17.0 Hz, 1H, =CH_2), 2.99 (s, 6H, N(CH_3)$_2$), 2.92–2.90 (m, 2H, CH_2), 2.35 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 150.9, 150.4, 143.1, 141.9 (Ar–C), 138.9 (=CH), 138.4 (=C), 133.2, 132.1, 128.9, 128.3, 128.1, 127.5 (Ar–C), 127.4 (=C), 120.6 (=CH$_2$), 117.5, 111.7 (Ar–C), 43.7 (CH$_2$), 40.3 (N(CH$_3$)$_2$), 21.6 (CH$_3$); HRMS (ESI), calcd for $C_{26}H_{28}NO_2S$ (MH$^+$), 418.1841, found, 418.1839; IR (KBr) 3055, 2976, 1607, 1595, 1443, 1358, 1311, 1145, 1087 cm^{-1}.

(Z)-1-(p-acetylphenyl)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4am)***

White solid; m.p. 128–129 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.95 (d, J = 8.4 Hz, 2H, Ar–H), 7.36–7.31 (m, 5H, Ar–H), 7.21 (d, J = 8.3 Hz, 2H, Ar–H), 7.17–7.15 (m, 2H, Ar–H), 7.06 (d, J = 7.8 Hz, 2H, Ar–H), 5.45–5.36 (m, 1H, =CH), 4.88 (dd, J = 1.2 Hz, 10.1 Hz, 1H, =CH_2), 4.68 (dd, J = 1.5 Hz, 18.5 Hz, 1H, =CH_2), 2.84 (d, J = 6.9 Hz, 2H, CH_2), 2.63 (s, 3H, CH_3), 2.34 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 197.7 (CO), 151.7, 143.9, 141.1, 138.7, 137.9 (=CH), 137.7 (=C), 137.0, 132.2, 131.5, 129.2, 128.2, 128.0, 127.9 (Ar–C), 127.7 (=C), 118.2 (=CH$_2$), 43.6 (CH$_2$), 26.8 (CH$_3$), 21.6 (CH$_3$); HRMS (ESI), calcd for $C_{26}H_{25}O_3S$ (MH$^+$), 417.1524, found, 417.1523; IR (KBr) 3049, 2975, 1685, 1631, 1600, 1441, 1361, 1302, 1149, 1086 cm^{-1}.

(E)-1-(p-acetylphenyl)-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene ***(4an)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.80 (d, J = 8.1 Hz, 2H, Ar–H), 7.46 (d, J = 7.8 Hz, 2H, Ar–H), 7.17 (d, J = 8.1 Hz, 2H, Ar–H), 7.07 (d, J = 7.8 Hz, 2H, Ar–H), 5.55–5.44 (m, 1H, =CH), 4.97 (d, J = 10.2 Hz, 1H, =CH_2), 4.84 (d, J = 16.8 Hz, 1H, =CH_2), 2.82 (t, J = 7.9 Hz, 2H, CH_2), 2.62 (m, 5H, CH_3, CH_2), 2.36 (s, 3H, CH_3), 1.67–1.50 (m, 2H, CH_2), 1.43–1.23 (m, 4H, 2CH_2), 0.91 (t, J = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 197.7 (CO), 154.6, 144.1 (Ar–C), 139.6 (=*C*), 138.4, 138.1 (Ar–*C*), 136.6 (=*C*H), 133.6, 131.6, 129.4, 128.0 (Ar–*C*), 127.9 (=*C*), 117.8 (=*C*H$_2$), 40.2 (*C*H$_2$), 32.2 (*C*H$_2$), 31.6 (*C*H$_2$), 29.1 (*C*H$_2$), 26.7 (*C*H$_3$), 22.5 (*C*H$_2$), 21.6 (*C*H$_3$), 14.1 (*C*H$_3$); HRMS (ESI), calcd for $C_{25}H_{31}O_3S$ (MH$^+$), 411.1994, found, 411.1994; IR (KBr) 2946, 1693, 1595, 1445, 1302, 1145, 1085 cm^{-1}.

*(Z)-1-(p-nitrilephenyl)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(4ao)***

White solid; m.p. 200–202 °C; ^{1}H NMR (300 MHz, CDCl$_3$) δ 7.66 (d, *J* = 8.4 Hz, 2H, Ar–*H*), 7.38–7.32 (m, 5H, Ar–*H*), 7.21–7.14 (m, 4H, Ar–*H*), 7.08 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 5.49–5.35 (m, 1H, =C*H*), 4.91 (dd, *J* = 1.4 Hz, 10.2 Hz, 1H, =C*H*$_2$), 4.69 (dd, *J* = 1.5 Hz, 18.3 Hz, 1H, =C*H*$_2$), 2.83 (d, *J* = 6.9 Hz, 2H, C*H*$_2$), 2.36 (s, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 152.1, 144.1, 140.6 (Ar–*C*), 138.7 (=*C*H), 137.6 (=*C*), 137.4, 132.1, 132.0, 131.9, 129.3, 128.3, 128.1, 128.0 (Ar–*C*), 127.8 (=*C*), 118.4 (=*C*H$_2$), 118.3 (*C*N), 112.7 (Ar–*C*), 43.6 (*C*H$_2$), 21.6 (*C*H$_3$); HRMS (ESI), calcd for $C_{25}H_{22}NO_2S$ (MH$^+$), 400.1371, found, 400.1368; IR (KBr) 3045, 2956, 2230, 1621, 1594, 1442, 1302, 1145, 1085 cm^{-1}.

*(E)-1-(p-nitrilephenyl)-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(4ap)***

Colorless oil; ^{1}H NMR (300 MHz, CDCl$_3$) δ 7.53–7.50 (m, 2H, Ar–*H*), 7.46–7.43 (m, 2H, Ar–*H*), 7.19 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.10–7.07 (m, 2H, Ar–*H*), 5.54–5.45 (m, 1H, =C*H*), 4.99 (dd, *J* = 1.2 Hz, 10.2 Hz, 1H, =C*H*$_2$), 4.83 (dd, *J* = 1.5 Hz, 18.6 Hz, 1H, =C*H*$_2$), 2.86–2.80 (m, 2H, C*H*$_2$), 2.56 (d, *J* = 6.3 Hz, 2H, C*H*$_2$), 2.39 (s, 3H, C*H*$_3$), 1.61–1.54 (m, 2H, C*H*$_2$), 1.41–1.34 (m, 4H, 2C*H*$_2$), 0.92 (t, *J* = 6.9 Hz, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 155.1, 144.3 (Ar–*C*), 139.6 (=*C*), 137.9 (=*C*H), 137.8, 133.3, 132.1, 131.8, 129.5 (Ar–*C*), 128.0 (=*C*), 118.4 (=*C*H$_2$), 117.9 (*C*N), 112.3 (Ar–*C*), 40.3 (*C*H$_2$), 32.2 (*C*H$_2$), 31.7 (*C*H$_2$), 29.1 (*C*H$_2$), 22.5 (*C*H$_2$), 21.6 (*C*H$_3$), 14.1 (*C*H$_3$); HRMS (ESI), calcd for $C_{24}H_{28}NO_2S$ (MH$^+$), 394.1841, found, 394.1841; IR (KBr) 3063, 2946, 2228, 1597, 1498, 1323, 1316, 1146, 1084 cm^{-1}.

*(Z)-1-(p-methylbenzoate)-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(4aq)***

White solid; m.p. 114–116 °C; ^{1}H NMR (300 MHz, CDCl$_3$) δ 7.98 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.31–7.12 (m, 9H, Ar–*H*), 7.03 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 5.43–5.29 (m, 1H, =C*H*), 4.85 (d, *J* = 9.9 Hz, 1H, =C*H*$_2$), 4.64 (d, *J* = 17.0 Hz, 1H, =C*H*$_2$), 3.90 (s, 3H, C*H*$_3$), 2.79 (d, *J* = 6.8 Hz, 2H, C*H*$_2$), 2.32 (s, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 166.7 (*C*O), 151.5, 143.8, 141.0 (Ar–*C*), 138.5 (=*C*H), 138.0 (=*C*), 137.6, 132.2, 131.4, 130.4, 129.4, 129.1, 128.4, 128.0, 127.9 (Ar–*C*), 127.7 (=*C*), 118.1 (=*C*H$_2$), 52.3 (O*C*H$_3$), 43.6 (*C*H$_2$), 21.6 (*C*H$_3$); HRMS (ESI), calcd for $C_{26}H_{25}O_4S$ (MH$^+$), 433.1474, found, 433.1477; IR (KBr) 2923, 1717, 1596, 1435, 1321, 1146 cm^{-1}.

*(E)-1-(p-methylbenzoate)-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(4ar)***

Colorless oil; ^{1}H NMR (300 MHz, CDCl$_3$) δ 7.89 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.46 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.17 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.05 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 5.57–5.46 (m, 1H, =C*H*), 4.98 (d, *J* = 10.1 Hz, 1H, =C*H*$_2$), 4.85 (dd, *J* = 1.3 Hz, 18.3 Hz, 1H, =C*H*$_2$), 3.91 (s, 3H, C*H*$_3$), 2.85 (t, *J* = 8.0 Hz, 2H, C*H*$_2$), 2.58 (d, *J* = 6.4 Hz, 2H, C*H*$_2$), 2.38 (s, 3H, C*H*$_3$), 1.63–1.50 (m, 2H, C*H*$_2$), 1.40–1.21 (m, 4H, 2C*H*$_2$), 0.93 (t, *J* = 6.9 Hz, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 166.7 (*C*O), 154.4, 144.0 (Ar–*C*), 139.4 (=*C*), 138.4 (=*C*H), 138.0, 133.7, 131.4, 129.9, 129.3, 129.2 (Ar–*C*), 128.1 (=*C*), 117.7 (=*C*H$_2$), 52.2 (*C*H$_3$), 40.3 (*C*H$_2$), 32.3 (*C*H$_2$), 31.6 (*C*H$_2$), 29.2 (*C*H$_2$), 22.5 (*C*H$_2$), 21.6 (*C*H$_3$), 14.1 (*C*H$_3$); HRMS (ESI), calcd for $C_{25}H_{31}O_4S$ (MH$^+$), 427.1943, found, 427.1942; IR (KBr) 2952, 2870, 1724, 1617, 1559, 1458, 1314, 1274, 1145, 1112 cm^{-1}.

*(Z)-1-(5-(2-chloropyridyl))-2-phenyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(4as)***

Yellow solid; m.p. 153–154 °C; ^{1}H NMR (300 MHz, CDCl$_3$) δ 8.07 (d, *J* = 2.1 Hz, 1H, Ar–*H*), 7.65 (dd, *J* = 2.4 Hz, 8.2 Hz, 1H, Ar–*H*), 7.37–7.31 (m, 4H, Ar–*H*), 7.19 (d, *J* = 8.2 Hz, 2H, Ar–*H*), 7.14–7.07 (m, 4H, Ar–*H*), 5.43–5.32 (m, 1H, =C*H*), 4.90 (d, *J* = 10.1 Hz, 1H, =C*H*$_2$), 4.71 (d, *J* = 17.0 Hz, 1H, =C*H*$_2$), 2.84 (d, *J* = 6.5 Hz, 2H, C*H*$_2$), 2.34 (s, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 153.6, 151.9, 151.1, 144.3 (Ar–*C*), 141.5 (=*C*), 137.7, 137.6 (Ar–*C*), 137.1 (=*C*H), 131.8, 129.4, 129.1, 128.9, 128.6, 128.2, 128.1, 127.9, 127.8 (Ar–*C*), 124.0 (=*C*), 118.4 (=*C*H$_2$), 43.7 (*C*H$_2$), 21.6 (*C*H$_3$); HRMS (ESI), calcd for $C_{23}H_{21}ClNO_2S$ (MH$^+$), 410.0982, found, 410.0977; IR (KBr) 3048, 2913, 1594, 1453, 1311, 1289, 1144 cm^{-1}.

*(E)-1-(5-(2-chloropyridyl))-2-pentyl-1-(p-tolylsulfonyl)-1,4-pentadiene **(4at)***

Colorless oil; ^{1}H NMR (300 MHz, CDCl$_3$) δ 7.80–7.79 (m, 1H, Ar–*H*), 7.48–7.42 (m, 3H, Ar–*H*), 7.28–7.22 (m, 3H, Ar–*H*), 5.60–5.47 (m, 1H, =C*H*), 5.01 (dd, *J* = 1.4 Hz, 10.1 Hz, 1H, =C*H*$_2$), 4.85 (dd, *J* = 1.5 Hz, 18.5 Hz, 1H, =C*H*$_2$), 2.93–2.80 (m, 2H, C*H*$_2$), 2.62 (t, *J* = 7.0 Hz, 2H, C*H*$_2$), 2.40 (s, 3H, C*H*$_3$), 1.64–1.51 (m, 2H, C*H*$_2$), 1.45–1.37 (m, 4H, 2C*H*$_2$), 0.93 (t, *J* = 6.9 Hz, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 156.8, 151.4, 151.1, 144.5 (Ar–*C*), 141.7 (=*C*), 137.5 (=*C*H), 135.0, 133.2, 129.7, 129.6, 127.9 (Ar–*C*), 123.8 (=*C*H$_2$), 118.0 (=*C*), 40.3 (*C*H$_2$), 32.2 (*C*H$_2$), 32.0 (*C*H$_2$), 29.2 (*C*H$_2$), 22.5 (*C*H$_2$), 21.6 (*C*H$_3$), 14.0 (*C*H$_3$); HRMS (ESI), calcd for $C_{22}H_{27}ClNO_2S$ (MH$^+$), 404.1451, found, 404.1448; IR (KBr) 2954, 2929, 1596, 1456, 1315, 1143, 1105, 1084 cm^{-1}.

*(Z)-2,3-diphenyl-1-(p-formylphenyl)-1-(p-tolylsulfonyl)propene **(4ba)***

White solid; m.p. 148–150 °C; ^{1}H NMR (300 MHz, CDCl$_3$) δ 10.10 (s, 1H, C*H*O), 7.94 (d, *J* = 7.5 Hz, 2H, Ar–*H*), 7.54 (d, *J* = 7.8 Hz, 2H, Ar–*H*), 7.28–6.97 (m, 12H, Ar–*H*), 6.67 (s, 2H, Ar–*H*), 3.48 (s, 2H, C*H*$_2$), 2.38 (s, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 191.4 (CHO), 152.0, 143.6 (Ar–*C*), 139.9 (=*C*), 137.0, 135.8, 135.5, 132.5, 131.9, 131.5, 129.2, 128.8, 128.7, 128.2, 128.0, 127.9, 127.8, 127.5, 127.3, 127.1 (Ar–*C*), 126.4 (=*C*), 44.5 (*C*H$_2$), 21.2 (*C*H$_3$); HRMS (EI), calcd for $C_{29}H_{24}O_3S$ (M$^+$), 452.1446, found, 452.1449. IR (KBr) 2837, 2732, 1732, 1699, 1595, 1454, 1384, 1145 cm^{-1}.

*(E)-2-benzyl-1-(p-formylphenyl)-1-(p-tolylsulfonyl)-1-heptene **(4bb)***

White solid; m.p. 75–77 °C; ^{1}H NMR (300 MHz, CDCl$_3$) δ 9.97 (s, 1H, C*H*O), 7.73 (d, *J* = 8.0 Hz, 2H, Ar–*H*), 7.50 (d, *J* = 8.1 Hz, 2H, Ar–*H*), 7.30–7.20 (m, 7H, Ar–*H*), 6.94 (d, *J* = 6.3 Hz, 2H, Ar–*H*), 3.24 (s, 2H, C*H*$_2$), 2.79 (t, *J* = 7.8 Hz, 2H, C*H*$_2$), 2.39 (s, 3H, C*H*$_3$), 1.67–1.47 (m, 2H, C*H*$_2$), 1.33–1.30 (m, 4H, 2C*H*$_2$), 0.89 (t, *J* = 6.7 Hz, 3H, C*H*$_3$); ^{13}C NMR (75 MHz, CDCl$_3$) δ 191.4 (CHO), 154.8, 143.9 (Ar–*C*), 140.9 (=*C*), 137.7, 137.1, 135.4, 131.8, 129.1, 129.0, 128.3, 128.2, 127.8, 127.7 (Ar–*C*), 126.4 (=*C*), 40.9 (*C*H$_2$), 31.8 (*C*H$_2$), 31.0 (*C*H$_2$), 28.9 (*C*H$_2$), 22.1 (*C*H$_2$), 21.3 (*C*H$_3$), 13.7 (*C*H$_3$); HRMS (EI), calcd for $C_{28}H_{30}O_3S$ (M$^+$), 446.1916, found, 446.1909; IR (KBr) 2949, 2856, 1740, 1618, 1599, 1495, 1467, 1383, 1142 cm^{-1}.

(Z)-5,6-diphenyl-4-(p-tolylsulfonyl)-1,4-hexadiene ***(4bc)***

White solid; m.p. 89–90 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.16–6.70 (m, 12H, Ar–*H*), 6.57 (d, J = 7.3 Hz, 2H, Ar–*H*), 5.99–5.90 (m, 1H, =C*H*), 5.22–5.13 (m, 2H, CH_2), 3.63 (s, 2H, CH_2), 3.55 (d, J = 5.8 Hz, 2H, CH_2), 2.24 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.4, 142.7, 139.0 (Ar–*C*), 138.1 (=*C*), 137.9 (=*C*H), 135.8, 134.1, 128.6, 128.4, 128.1, 127.3, 127.0, 126.9 (Ar–*C*), 126.3 (=*C*), 116.6 (=*C*H_2), 43.1 (*C*H_2), 32.9 (*C*H_2), 21.2 (*C*H_3); HRMS (EI), calcd for $C_{25}H_{24}O_2S$ (M^+), 388.1497, found, 388.1492; IR (KBr) 1645, 1595, 1489, 1310, 1150 cm^{-1}.

(Z)-1,2,4-triphenyl-3-(p-tolylsulfonyl)-2-butene ***(4bd)***

White solid; m.p. 136–138 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.39–7.27 (m, 6H, Ar–*H*), 7.13–6.93 (m, 9H, Ar–*H*), 6.76–6.73 (m, 4H, Ar–*H*), 4.30 (s, 2H, CH_2), 3.76 (s, 2H, CH_2), 2.32 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 152.1, 142.8, 140.1 (Ar–*C*), 138.1 (=*C*), 137.9, 137.8, 135.6, 128.6, 128.5, 128.4, 128.3, 128.2, 128.0, 127.3, 127.0, 126.9, 126.4 (Ar–*C*), 126.3 (=*C*), 43.9 (*C*H_2), 34.4 (*C*H_2), 21.2 (*C*H_3); HRMS (EI), calcd for $C_{29}H_{26}O_2S$ (M^+), 438.1654, found, 438.1660; IR (KBr) 2916, 1624, 1597, 1495, 1452, 1300, 1144 cm^{-1}.

(E)-1-phenyl-3-benzyl-1-(p-tolylsulfonyl)-2-octene ***(4be)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.52 (d, J = 8.1 Hz, 2H, Ar–*H*), 7.21–7.09 (m, 10H, Ar–*H*), 6.87 (d, J = 6.9 Hz, 2H, Ar–*H*), 3.84 (s, 2H, CH_2), 3.45 (s, 2H, CH_2), 2.48 (t, J = 7.5 Hz, 2H, CH_2), 2.31 (s, 3H, CH_3), 1.24–1.11 (m, 6H, 3CH_2), 0.75 (t, J = 6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 155.0, 143.3, 139.2 (Ar–*C*), 137.9 (=*C*), 137.2, 136.9, 129.1, 128.3, 128.2, 127.9, 127.8, 127.0, 126.3 (Ar–*C*), 126.0 (=*C*), 39.6 (*C*H_2), 34.9 (*C*H_2), 32.8 (*C*H_2), 31.8 (*C*H_2), 28.2 (*C*H_2), 22.1 (*C*H_2), 21.2 (*C*H_3), 13.6 (*C*H_3); HRMS (EI) calcd for $C_{28}H_{32}O_2S$ (M^+), 432.2123, found, 432.2117. IR (film) 2960, 2858, 1641, 1601, 1452, 1309, 1141 cm^{-1}.

(Z)-5-phenyl-4-(p-tolylsulfonyl)-1,4-nonadiene ***(4ca)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.34–7.08 (m, 5H, Ar–*H*), 6.97 (d, J = 8.0 Hz, 2H, Ar–*H*), 6.82 (d, J = 6.9 Hz, 2H, Ar–*H*), 5.95–5.86 (m, 1H, =C*H*), 5.23–5.13 (m, 2H, CH_2), 3.45 (d, J = 5.8 Hz, 2H, CH_2), 2.34–2.29 (m, 5H, CH_2, CH_3), 1.21–1.19 (m, 4H, 2CH_2), 0.77 (t, J = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 153.9, 142.5 (Ar–*C*), 138.4 (=*C*), 138.3 (=*C*H), 137.3, 134.3, 128.5, 128.2, 127.2, 127.0 (Ar–*C*), 126.9 (=*C*), 116.1 (=*C*H_2), 37.3 (*C*H_2), 32.4 (*C*H_2), 28.9 (*C*H_2), 22.3 (*C*H_2), 21.1 (*C*H_3), 13.5 (*C*H_3); HRMS (EI), calcd for $C_{22}H_{26}O_2S$ (M^+), 354.1654, found, 354.1649; IR (film) 2958, 2862, 2181, 1640, 1597, 1504, 1491, 1300, 1146 cm^{-1}.

(Z)-5-butyl-4-(p-tolylsulfonyl)-1,4-decadiene ***(4cb)***

Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.72 (d, J = 7.9 Hz, 2H, Ar–*H*), 7.27 (d, J = 7.8 Hz, 2H, Ar–*H*), 5.82–5.73 (m, 1H, =C*H*), 5.08–5.03 (m, 2H, CH_2), 3.20 (d, J = 5.2 Hz, 2H, CH_2), 2.51–2.41 (m, 5H, CH_2, CH_3), 2.16–2.10 (m, 2H, CH_2), 1.32–1.22 (m, 10H, 5CH_2), 0.91–0.82 (m, 6H, 2CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.2 (=*C*), 143.1, 139.6 (Ar–*C*), 134.6 (=*C*H), 133.4, 129.1 (Ar–*C*), 126.8 (=*C*), 115.7 (=*C*H_2), 33.6 (*C*H_2), 32.9 (*C*H_2), 32.5 (*C*H_2), 31.8 (*C*H_2), 29.8 (*C*H_2), 28.2 (*C*H_2), 22.6 (*C*H_2), 22.1 (*C*H_2), 21.1 (*C*H_3), 13.6 (*C*H_3), 13.5 (*C*H_3); HRMS (EI), calcd for $C_{21}H_{32}O_2S$ (M^+), 348.2123, found, 348.2117; IR (film) 2957, 2861, 1599, 1494, 1458, 1182, 1144 cm^{-1}.

(Z)-2-phenyl-1-(p-nitrilephenyl)-1-(p-tolylsulfonyl)-1-hexene ***(4cc)***

White solid; m.p. 127–129 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.68 (d, J = 8.0 Hz, 2H, Ar–*H*), 7.39–7.37 (m, 5H, Ar–*H*), 7.22–7.08 (m, 6H, Ar–*H*), 2.38 (s, 3H, CH_3), 2.09 (t, J = 7.5 Hz, 2H, CH_2), 1.12–1.03 (m, 4H, 2CH_2), 0.64 (t, J = 6.7 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 154.8 (=*C*), 143.6, 139.3, 138.7, 137.3, 137.2, 131.6, 128.8, 127.8, 127.6, 127.5 (Ar–*C*), 127.4 (=*C*), 118.1 (*C*N), 112.1 (Ar–*C*), 38.3 (*C*H_2), 28.9 (*C*H_2), 21.9 (*C*H_2), 21.2 (*C*H_3), 13.2 (*C*H_3); HRMS (EI), calcd for $C_{26}H_{25}NO_2S$ (M^+), 415.1606, found, 415.1608; IR (KBr) 2956, 2226, 1624, 1597, 1441, 1313, 1146, 1084 cm^{-1}.

(Z)-3-butyl-1-(p-nitrilephenyl)-1-(p-tolylsulfonyl)-1-heptene ***(4cd)***

White solid; m.p. 84–86 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.51 (d, J = 8.0 Hz, 2H, Ar–*H*), 7.43 (d, J = 8.0 Hz, 2H, Ar–*H*), 7.17 (d, J = 7.9 Hz, 2H, Ar–*H*), 7.07 (d, J = 8.0 Hz, 2H, Ar–*H*), 2.80 (t, J = 7.3 Hz, 2H, CH_2), 2.36 (s, 3H, CH_3), 1.78 (t, J = 7.3 Hz, 2H, CH_2), 1.71–1.46 (m, 2H, CH_2), 1.42–1.29 (m, 4H, 2CH_2), 1.27–1.17 (m, 2H, CH_2), 1.07–0.98 (m, 2H, CH_2), 0.90 (t, J = 7.1 Hz, 3H, CH_3), 0.66 (t, J = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 157.9 (=*C*), 143.8, 139.6, 137.7, 136.4, 131.9, 131.3, 129.1 (Ar–*C*), 127.5 (=*C*), 118.1 (*C*N), 111.6 (Ar–*C*), 35.2 (*C*H_2), 31.9 (*C*H_2), 31.2 (*C*H_2), 29.8 (*C*H_2), 28.8 (*C*H_2), 22.2 (*C*H_2), 22.1 (*C*H_2), 21.2 (*C*H_3), 13.7 (*C*H_3), 13.2 (*C*H_3); MS (EI), 381.2 (M^+ − CN); anal. calcd for $C_{25}H_{31}NO_2S$, C, 73.31; H, 7.63; N, 3.42. found: C, 73.04; H, 7.54; N, 3.12. IR (KBr) 2957, 2930, 2226, 1597, 1493, 1382, 1294, 1182, 1016 cm^{-1}.

(E)-1-phenyl-1-(p-fluorophenyl)-2-(p-tolylsulfonyl)-1,4-pentadiene ***(4da)***

White solid; m.p. 134–135 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.31–6.94 (m, 13H, Ar–*H*), 6.19–5.96 (m, 1H, =C*H*), 5.15–5.03 (m, 2H, =CH_2), 3.37 (d, J = 6.2 Hz, 2H, CH_2), 2.37 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 164.2, 161.3, 151.4, 142.9, 141.1 (Ar–*C*), 138.6(=*C*H), 134.8 (=*C*), 129.4, 129.3, 128.7, 128.6, 127.5 (Ar–*C*), 127.3(=*C*), 116.8 (=*C*H_2), 115.3, 115.0 (Ar–*C*), 34.6 (*C*H_2), 21.3 (*C*H_3); HRMS (EI), calcd for $C_{24}H_{21}FO_2S$ (M^+), 392.1246, found: 392.1243; IR (KBr) 2953, 2872, 1504, 2349, 1146, 1084 cm^{-1}.

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